

llmXive Automated Review of ABot-AgentOS: A General Robotic Agent OS with Lifelong Multi-modal Memory

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a catastrophic failure in claim accuracy due to the apparent concatenation of three distinct, mutually exclusive drafts (ABot-AgentOS, SparkVLA-M1, and a Reward Engine fragment) into a single manuscript. This results in multiple fatal contradictions where the same system is claimed to achieve different, incompatible results on the same benchmarks, and where the text cites models that do not exist.

First, the manuscript cites and compares against non-existent baselines. Tables 1, 2, and 3 list ‘GPT-5.4 mini’, ‘Gemini 2.5 Pro’, and ‘Qwen3.6-Plus’ as baselines or models. As of the current date, GPT-5 and Gemini 2.5 have not been released, and Qwen3.6 is not a public model. The paper claims these models achieved specific scores (e.g., 93.1% on LoCoMo for GPT-5.4), but since the models do not exist, these results are fabricated or hallucinated. This invalidates the comparative claims in the memory evaluation section.

Second, the results are internally inconsistent due to the merged drafts. The Introduction (Section 1) claims a 98.6% success rate on LIBERO and 80.5% on LIBERO-Plus for ABot-AgentOS. However, the ‘Experiments’ section (Section 5) reports results for ABot-AgentOS on ‘EmbodiedWorldBench’ with TSR of 61.96% and 68.79%. Furthermore, a separate fragment (Section 5, ‘SparkVLA-M1’) claims 77.5% on LIBERO-Plus. The paper fails to distinguish between these different systems or benchmarks, presenting contradictory numbers for the same task (LIBERO-Plus: 80.5% vs 77.5%) without explanation.

Third, the citation keys in the text (e.g., \cite{yang2026abot}) do not match the bibliography entries (e.g., huo2026abot), breaking the link between the claim of skill capabilities and the supporting evidence.

Finally, the claim of “99% accuracy” for privacy gating in Section 2.4 is a specific quantitative assertion with no corresponding table or experimental description in the provided text, rendering it an unsupported load-bearing claim.

The paper cannot be accepted in its current form as the core scientific claims are unsupported by the provided evidence or are factually impossible due to the use of non-existent models.

Action items.

- **[fatal]** The paper presents a catastrophic failure in claim accuracy due to the apparent concatenation of three distinct, mutually exclusive drafts (ABot-AgentOS, SparkVLA-M1, and a Reward Engine fragment) into a single manuscript. This results in multiple fatal contradictions where the same system is claimed to achieve different, incompatible results on the same benchmarks, and where the text cites models that do not exist. First, the manuscript cites and compares against non-existent baselines. Tables

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a decorative logo with a placeholder caption ('xxx') and does not contain any scientific data or information relevant to the paper's claims.

Figure 2

The figure provides a clear, high-level overview of the system architecture with well-defined input/output flows and component roles. The visual hierarchy effectively distinguishes between edge and cloud components, and the caption accurately describes the diagram's content.

Figure 3

Figure 3 is a clear and well-structured schematic that effectively visualizes the Agent Harness workflow described in the caption. All components (LLM, Skill Runner, Verifier) and their interactions (inputs, outputs, feedback loops) are explicitly labeled and visually distinct.

Figure 4

The figure provides a clear visual overview of the memory architecture but fails to visually connect the 'Self-Evo Subagents' to the rest of the system, leaving the 'gated runtime evo-assets' mentioned in the caption unrepresented in the data flow.

Figure 5

Figure 5 effectively illustrates the system's self-evolution mechanism with two concrete examples (visual and temporal memory). The layout is clear, the flow from failure to asset generation is logical, and the caption accurately describes the visual content.

Figure 6

The figure provides a visual example of a task within the benchmark environment but fails to visually represent the '16 scenes' and 'four difficulty levels' mentioned in the caption, making the overview incomplete.

Figure 7

The figure is a composite of three panels (a, b, c) that are not clearly unified by the caption, which only describes the general training pipeline. Panel (c) shows a map example without context, and panel (b) lacks detailed axis labels or legends for the advantage diagrams.

Figure 8

The figure provides a clear visual overview of the sequential split process and state transfer, but it contains a spelling error in the ‘accumulated evo state’ label and omits visual representation of the failure gating mechanism described in the caption.

Figure 9

The figure effectively visualizes performance gains across benchmarks, but the x-axis label is redundant and the specific sub-category acronyms in the Mem-Gallery section are undefined.

Figure 10

The figure provides a clear visual overview of the reward engine architecture, but the caption is truncated mid-sentence and appears to be a duplicate of the caption for Figure 11.

Figure 11

The figure is a clear and well-structured diagram that effectively visualizes the three-module reward engine described in the text. However, the caption is truncated and appears to be a duplicate of the caption for Figure 10.

Action items.

- **[fatal]** Figure 1: The caption is a placeholder (‘xxx’) and the image is a decorative logo rather than a scientific figure, failing to provide any data, system overview, or experimental results.
- **[writing]** Figure 2: The ‘Agent Harness’ block lists ‘Skill Evolvment’, but the caption describes ‘skill evolvment’ as a function managed by the harness; the diagram implies it is a static component rather than a process, which is slightly ambiguous.
- **[writing]** Figure 2: The ‘Skills & Tools’ block contains an ellipsis (‘...’) without a legend or caption note explaining what other skills are implied, which is acceptable but could be more specific.
- **[science]** Figure 4: The ‘Self-Evo Subagents’ block (Diagnoser, Hypothesis, Compiler, Gate Analyst) is visually disconnected from the ‘Agent Controller’ and ‘Memory DB’ components. The caption describes ‘gated runtime evo-assets’ and ‘diagnosed’ traces, but the diagram lacks arrows showing how the ‘Failure Trace’ enters the subagents or how the resulting ‘evo-assets’ are fed back into the system (e.g., to the Skills or Controller).
- **[writing]** Figure 4: The purple dashed arrow labeled ‘self-evo feedback’ originates from

the ‘Self-Evo Subagents’ block but points to the ‘Perception’ module. This contradicts the caption’s description of converting failure traces into ‘runtime evo-assets’ for ‘later deployments,’ as the visual flow suggests a direct feedback loop to perception rather than a system-level update.

- **[science]** Figure 6: The caption claims the benchmark covers ‘16 scenes’ and ‘four difficulty levels’, but the figure only displays a single map view without any visual indicators (e.g., a legend, color coding, or inset grid) to distinguish these levels or show the other scenes.
- **[writing]** Figure 6: The central text box describes a specific task (‘Check if Tom is on the street...’), but the figure lacks a title or label explicitly identifying this as the ‘Embodied-WorldBench’ example or task.
- **[writing]** Figure 7: The caption describes the figure as an ‘Overview of the training pipeline,’ but the image is a composite of three distinct panels labeled (a), (b), and (c) with no unifying title or explanation of how they relate to each other in the caption.
- **[writing]** Figure 7: Panel (c) is titled ‘Trajectory Example’ and shows a map, but the caption does not mention this visual example or explain its relevance to the training pipeline described.
- **[science]** Figure 7: Panel (b) ‘Reinforcement Learning’ shows ‘Episode advantage’ and ‘Step advantage’ diagrams but lacks numerical axes, units, or a clear legend explaining the specific meaning of the colored lines and nodes beyond the labels.
- **[writing]** Figure 8: The label ‘accumulated evo state’ contains a spelling error (‘accumulated’ is misspelled as ‘accumulated’).
- **[science]** Figure 8: The diagram depicts a linear progression of ‘Self-Evo Subagents’ and ‘Evo’ states, but the caption describes a process of ‘diagnosing and gating’ failures; the figure lacks visual indicators (e.g., rejection paths or gating mechanisms) to illustrate how failures are filtered out.
- **[writing]** Figure 9: The x-axis label ‘Absolute gain Δ (+ Self-evo - Static)’ is syntactically confusing; the delta symbol and the parenthetical formula are redundant and should be simplified to ‘Absolute gain (Self-evo - Static)’.
- **[writing]** Figure 9: The y-axis labels for the ‘Mem-Gallery’ section (e.g., ‘FR’, ‘VS’, ‘TTL’) are undefined acronyms; the caption or figure should expand these to full category names for clarity.
- **[writing]** Figure 10: The caption text is truncated mid-sentence at the end (‘validates each update before’), failing to complete the description of Module 3.
- **[writing]** Figure 10: The caption appears to be a duplicate of Figure 11’s caption (identical text and filename), suggesting a copy-paste error in the manuscript.
- **[writing]** Figure 11: The caption text is truncated at the end (‘validates each update before’), cutting off the sentence and the figure filename.

- **[writing]** Figure 11: The caption is a duplicate of the caption provided for Figure 10, despite the figure content being identical to Figure 10.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper generally maintains a high level of technical precision, but it relies heavily on subfield-specific acronyms and shorthand that are undefined at their first occurrence, creating barriers for a competent reader from an adjacent field (e.g., a robotics systems engineer or a general NLP researcher).

The most significant omissions involve the term “evo-assets” in Section 2.2. This is a coined term for the paper’s specific mechanism of self-evolution, yet it is introduced without any explanation of what an “asset” constitutes in this context (e.g., code snippets, prompt templates, or graph rules). Similarly, “ReAct” is used in Section 2.1 without expansion; while standard in the LLM agent community, it is not universal knowledge across all of AI.

Additionally, the paper introduces “GiGPO” in Section 3.1 without spelling out the acronym, forcing the reader to guess the meaning of the algorithm based on the surrounding text. The use of “NavMesh” and “JSON DSL” also assumes familiarity with game-engine terminology and specific software engineering concepts that may not be immediate to a pure machine learning theorist.

Finally, while “VLM” and “VLA” are foundational to the paper’s topic, they appear as acronyms before being explicitly defined in the text. A strict adherence to the “define at first use” rule would improve accessibility for the target “adjacent-field PhD” audience. These are minor text edits that would significantly lower the cognitive load for non-specialists without diluting the technical content.

Action items.

- **[writing]** The paper generally maintains a high level of technical precision, but it relies heavily on subfield-specific acronyms and shorthand that are undefined at their first occurrence, creating barriers for a competent reader from an adjacent field (e.g., a robotics systems engineer or a general NLP researcher). The most significant omissions involve the term “evo-assets” in Section 2.2. This is a coined term for the paper’s specific mechanism of self-evolution, yet it is introduced without any explan

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper suffers from a catastrophic logical breakdown due to the apparent concatenation of three distinct, mutually exclusive research drafts into a single manuscript. The argument structure is fractured, rendering the central thesis incoherent.

1. Contradictory Core Premises (Fatal): The manuscript contains three separate “Introduction” sections (labeled e000, e001, e002 in the source chunks) that propose fundamentally different systems:

- **e000:** Introduces **VLA FM**, a VLA foundation model integrating 6M trajectories, focusing on “Action Manifold Learning” and reporting 98.6% success on LIBERO.
- **e001:** Introduces **ABot-AgentOS**, a general robotic agent OS with a “Verification-aware ReAct” loop and “Lifelong Multi-modal Memory,” evaluated on “EmbodiedWorldBench.”
- **e002:** Introduces **SparkVLA-M1**, a cross-embodiment framework using “JAT” and “Long-Cat,” also reporting 98.6% on LIBERO but with different architectural details (Qwen3-VL vs. Qwen3.6-Plus).

The **Conclusion** (e002) exclusively summarizes **ABot-AgentOS**, completely ignoring the VLA FM and SparkVLA-M1 contributions. This creates a fatal logical gap: the premises (the existence of three different systems) do not support the conclusion (a single unified system). The reader cannot determine which system the results actually validate.

2. Non-Entailed Conclusions and Mismatched Evidence:

- The **Experiments** section (e001) presents tables for “ABot-AgentOS” (using Qwen3.6-Plus) but the text often references “VLA FM” or “SparkVLA-M1” results (e.g., the 98.6% LIBERO score). The conclusion that “ABot-AgentOS functions as both an online reasoning-execution layer and a persistent learning substrate” is not entailed by the data if the data actually belongs to a different VLA model (VLA FM/SparkVLA-M1).
- The **Memory Evaluation** claims (e.g., “ABot-AgentOS Static obtains 87.5 overall”) are presented as evidence for the “Agent OS” thesis, but the “Introduction” (e000) frames the paper as a “VLA foundation model” paper. The logical link between the memory benchmarks and the VLA architecture is severed.

3. Scope Inflation and Definition Drift:

- The term “ABot-AgentOS” is used to describe a system with a “Tiny LLM” and “Large LLM” (e000) in one section, but the “Model Training” section (e001) describes training a “student policy” via RL on a “text-based environment,” which contradicts the “Agent OS” runtime description.
- The dataset size is stated as “over 6 million trajectories” in e000 and e001, but e002 mentions “6M+ trajectories” with a different cleaning pipeline (“CleanCore”). The inconsistency in data processing definitions undermines the validity of the reported performance gains.

Fix Required: The authors must select **one** coherent research narrative (either the VLA foundation model, the Agent OS, or the cross-embodiment framework) and remove the contradictory sections. The Introduction, Method, Experiments, and Conclusion must all refer to the *same* system with consistent naming, architecture, and evaluation metrics. Currently, the paper argues for three different things simultaneously, making the logical chain broken.

Action items.

- **[fatal]** The paper presents three distinct, mutually exclusive ‘Introduction’ sections (e000, e001, e002) proposing different core systems: ‘VLAFM’ (VLA foundation model), ‘ABot-AgentOS’ (agent OS), and ‘SparkVLA-M1’ (cross-embodiment VLA). The Conclusion (e002) only summarizes ABot-AgentOS, rendering the VLAFM and SparkVLA-M1 results (e000, e001) logically orphaned and unsupported by the paper’s final thesis.
- **[fatal]** Section ‘Model Training’ (e001) describes a pipeline for ‘ABot-AgentOS’ using ‘Qwen3.6-Plus’, while the ‘Introduction’ (e000) and ‘Experiments’ (e001) describe ‘VLAFM’ and ‘SparkVLA-M1’ using ‘Qwen3-VL’ and ‘DiT’ backbones. The results tables (e001) report metrics for ‘ABot-AgentOS’ but the text claims these validate the ‘VLAFM’ or ‘SparkVLA-M1’ hypotheses, creating a non-sequitur between the proposed method and the reported evidence.
- **[science]** The ‘Introduction’ (e000) claims ‘VLAFM’ achieves 98.6% on LIBERO, while the ‘Experiments’ section (e001) reports 98.6% for ‘ABot-AgentOS’ on a subset of Embodied-WorldBench. The paper fails to reconcile whether these are the same system or different systems, and the ‘Conclusion’ (e002) ignores the LIBERO results entirely, breaking the logical chain from premise to conclusion.
- **[writing]** The ‘Memory Evaluation’ section (e001) claims ‘ABot-AgentOS Static’ outperforms ‘Mem0’ on LoCoMo (87.5 vs 85.6), but the ‘Introduction’ (e000) and ‘Conclusion’ (e002) frame the paper’s contribution as a ‘VLA foundation model’ or ‘Agent OS’ without clarifying if the memory results apply to the VLA or the OS, creating ambiguity in the scope of the claimed contribution.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper exhibits significant overreach regarding the scope of its validation versus its claims of generality and physical deployment. The title “ABot-AgentOS: A General Robotic Agent OS” and the introduction’s promise of “one mind, multiple forms” operating across “heterogeneous platforms” (humanoids, quadrupeds) create an expectation of broad, real-world applicability. However, the experimental evidence (Section 5) is strictly confined to simulated environments (UnrealZoo) and text-based sandboxes. The claim that the system has been “validated on real

and simulated platforms” is not supported by any data in the results section; no physical robot trials are reported. This is a structural overreach where the framing (Title/Intro/Conclusion) asserts a capability (general physical operation) that the evidence (simulation only) does not license.

Furthermore, the claim of “physical execution” in the conclusion is contradicted by the methodology, which relies on “text-based environment construction” and “sandboxed trajectory distillation.” While the architecture is designed for physical robots, the paper presents the results as if the system has already bridged the sim-to-real gap, which it has not demonstrated. The privacy gating claim of “over 99% accuracy” also lacks necessary context regarding the test set, potentially overstating the robustness of the privacy mechanism. To resolve this, the authors must either narrow their claims to “simulated generalization” and “framework for physical execution” or provide the missing real-world experimental data to support the current broad assertions.

Action items.

- **[science]** Title/Abstract claim ‘General Robotic Agent OS’ across ‘heterogeneous platforms’ (humanoids, quadrupeds), but Section 5 results are exclusively simulated (UnrealZoo). No physical robot data exists. Replace ‘General’ with ‘Simulated’ or add real-world validation.
- **[science]** Conclusion states validation on ‘real and simulated platforms (quadruped, humanoid),’ yet Section 5 reports only simulation metrics. Remove ‘real’ validation claims or provide missing physical robot experimental evidence.
- **[writing]** Conclusion claims connection to ‘physical execution,’ but Section 4/5 rely on ‘text-based environments’ and ‘sandboxes.’ Rephrase to clarify the system is designed for physical use but currently validated only in simulation.
- **[writing]** Section 3.2 claims ‘over 99% accuracy’ for privacy gating without specifying the test dataset or conditions. Clarify the scope of this metric to avoid implying universal robustness across all potential inputs.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper describes a robotic agent operating system (ABot-AgentOS) and a benchmark (EmbodiedWorldBench) evaluated primarily in simulation (UnrealZoo) and on public, pre-existing datasets (LoCoMo, OpenEQA, etc.). The methodology does not involve the collection of new human-subject data, the release of personally identifiable information (PII), or the use of scraped data in a manner that appears to violate terms of service (the training data cited are standard open-source robotics datasets like OXE and RoboCoin).

The system includes a “Privacy-aware Gating” mechanism for memory sharing, explicitly

designed to filter out PII (faces, names, personal objects) before cloud synchronization. While the paper does not provide a detailed audit of the effectiveness of this filter beyond a claimed 99% accuracy, the inclusion of this safeguard discussion is appropriate for the scope of the work. The “LLM-as-Judge” components are used for internal reward generation and evaluation within the training loop, not for external surveillance or deceptive interaction with humans.

There are no dual-use capabilities described that meaningfully lower the barrier to specific harms (e.g., automated vulnerability exploitation, biological synthesis, or targeted disinformation) beyond the general capabilities of the underlying foundation models, which are not the novel contribution of this paper. The research is low-risk by construction, and the paper adequately addresses the relevant safety considerations for a simulation-based robotics framework. No specific safety or ethics disclosures are missing.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a comprehensive system architecture and extensive benchmarking results, but the evidentiary design in the experimental section has significant gaps that prevent the claims from being fully substantiated.

First, the primary agent evaluation (Table 1) reports a substantial improvement (11.99% TSR) over a ReAct baseline using the same underlying LLM. However, the results are presented as single-point estimates without any indication of statistical stability (e.g., standard deviation, confidence intervals, or seed count). In embodied agent benchmarks with ~200 tasks, a single run can easily fluctuate by several percentage points due to stochasticity in the environment or the LLM. Without reporting results across multiple seeds (e.g., 3-5), the reader cannot distinguish a genuine architectural effect from a lucky seed or sampling noise.

Second, the memory evaluation section compares the proposed “Static” graph memory against various baselines (MemGPT, Mem0, etc.). The experimental setup states the “base hybrid graph retriever is fixed,” but it is unclear if the baselines were evaluated with their optimal configurations or if they were given the same LLM backbone and prompt engineering resources. If the proposed method benefited from extensive prompt tuning or a more powerful LLM for the “Writer/Answerer” components while the baselines were run in a “zero-shot” or default mode, the performance gap may reflect resource asymmetry rather than the efficacy of the graph memory schema. The paper needs to explicitly confirm that all baselines were given a fair, comparable opportunity for tuning.

Third, the “Lifelong Self-Evolution” results (Figure 4) demonstrate incremental gains across splits. While the mechanism is described, the evidence lacks a measure of variance. The gains (e.g., +0.4 on Mem-Gallery, +4.1 on NExT-QA) are relatively small. Without reporting the standard error or running the evolution protocol on multiple random seeds/splits, it is impossible

to determine if these gains are statistically significant or merely artifacts of the specific data ordering.

Finally, the claim of “99% accuracy” in the privacy-aware gating module (Section 2.4) is presented without supporting data on the test set size or class balance. A binary classification claim of this magnitude requires a clear definition of the evaluation set (e.g., number of items, ratio of private to public) to ensure the result is not trivial or overfitted to a specific subset.

To resolve these issues, the authors should re-run the main agent experiments with multiple seeds and report variance, clarify the tuning parity for all baselines in the memory section, and provide statistical significance testing or error bars for the self-evolution gains.

Action items.

- **[science]** The paper presents a comprehensive system architecture and extensive benchmarking results, but the evidentiary design in the experimental section has significant gaps that prevent the claims from being fully substantiated. First, the primary agent evaluation (Table 1) reports a substantial improvement (11.99% TSR) over a ReAct baseline using the same underlying LLM. However, the results are presented as single-point estimates without any indication of statistical stability (e.g., standard deviat

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment of the quantitative results in this manuscript is insufficient to support the claimed improvements and comparative rankings. The primary issue is the systematic absence of uncertainty quantification. In Sections 4.1 and 4.2, performance metrics (TSR, GCR, and various QA accuracies) are reported as precise point estimates (e.g., “61.96%”, “87.5”) with no accompanying standard deviation, standard error, or confidence intervals. Given that the experiments involve stochastic elements (LLM sampling, RL training, and random task generation), reporting a single number or a mean without spread implies a false precision that the data cannot support. For instance, Table 1 claims an 11.99% improvement in TSR; without knowing the variance across seeds, it is impossible to determine if this difference is statistically significant or within the noise floor of the baseline.

Furthermore, the paper engages in extensive multiple hypothesis testing without correction. The memory evaluation compares the proposed method against numerous baselines across five distinct benchmarks (LoCoMo, OpenEQA, Mem-Gallery, NExT-QA, EgoLife), resulting in dozens of pairwise comparisons. Highlighting the “best” scores in these tables without applying a correction for multiple comparisons (such as Bonferroni or Benjamini-Hochberg) inflates the family-wise error rate, making it highly probable that some reported “state-of-the-art” results are statistical flukes.

Finally, the “Lifelong Self-Evolution” results present specific gain values (e.g., “+4.1 points”)

derived from a sequential split process. The statistical validity of these gains is unverified because there is no report of variance across independent evolution runs or splits. The claim that the system “improves” relies on a single trajectory of self-evolution, which could be driven by the specific random seed of the initial split rather than the robustness of the method. To be scientifically rigorous, the authors must re-run experiments with multiple seeds to generate variance estimates and apply appropriate corrections for the multiplicity of comparisons made across the benchmarks.

Action items.

- **[science]** Table 1 (Agent Evaluation) reports TSR/GCR improvements (e.g., +11.99%) with 2 decimal places but provides no uncertainty metrics (SD, SE, or CI) or seed count. In stochastic RL/agent tasks, a single run is insufficient to claim ‘improvement.’ Report mean $\pm$ SD over 3 seeds or explicitly state the result is from a single run and remove implied precision.
- **[science]** Tables 2-6 (Memory Benchmarks) compare ABot-AgentOS against ~15 baselines across 5 datasets (75+ pairwise comparisons) without any multiple-comparison correction (e.g., Bonferroni, Holm, or FDR). With this volume of tests, several ‘best’ results are likely false positives. Apply a correction method or explicitly acknowledge the uncorrected multiplicity risk.
- **[science]** Section ‘Lifelong Self-Evolution Results’ reports specific point gains (e.g., ‘NExT-QA improves by 4.1 points’) derived from a sequential split protocol. No variance across splits or confidence intervals are provided to distinguish genuine evolution from random fluctuation in the gate selection process. Report the distribution of gains across multiple independent evolution runs.
- **[writing]** The ‘Edge-Cloud Collaborative Memory Management’ section claims ‘over 99% accuracy’ in identifying shareable items based on a single metric without reporting the sample size (N), confidence interval, or the specific test set composition. A point estimate of 99% is statistically meaningless without the denominator and error bounds.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript suffers from severe structural fragmentation, likely due to the concatenation of multiple draft versions or distinct project descriptions into a single file. The most critical readability failure is the presence of three mutually exclusive “Introduction” sections (e002) describing different systems: “ABot-AgentOS,” “VLA FM,” and “SparkVLA-M1.” A reader cannot determine the paper’s primary contribution because the opening section contradicts itself immediately. This is not a stylistic issue but a fundamental breakdown in the document’s

narrative thread.

Similarly, the body of the paper repeats content in conflicting ways. Section 3 and Section 4 both contain detailed descriptions of reward engines and training pipelines that overlap significantly but use different terminology and section headers. This forces the reader to re-read and cross-reference to understand if these are distinct methods or redundant descriptions. The flow is further disrupted by the abrupt insertion of “Conclusion” text in the middle of the document (e002) before the “Introduction” is even finished.

While the individual sentences within the coherent fragments are generally grammatically sound, the global organization prevents the reader from moving through the argument without constant confusion. The paper currently reads as a collection of disjointed drafts rather than a unified scientific report. To pass, the authors must perform a rigorous structural edit: select one primary system narrative, remove all contradictory duplicate sections, and ensure the Introduction, Methods, Experiments, and Conclusion tell a single, consistent story.

Action items.

- **[science]** The manuscript suffers from severe structural fragmentation, likely due to the concatenation of multiple draft versions or distinct project descriptions into a single file. The most critical readability failure is the presence of three mutually exclusive “Introduction” sections (e002) describing different systems: “ABot-AgentOS,” “VLA FM,” and “SparkVLA-M1.” A reader cannot determine the paper’s primary contribution because the opening section contradicts itself immediately. This is not a stylist